

TRACELESS $SU(2)$ CHARACTERS AND $\mathbb{Z}/4$ INSTANTON GRADINGS FOR TWO-BRIDGE AND $(3, n)$ -TORUS KNOTS

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ABSTRACT. We collect the generators, the $\mathbb{Z}/4$ gradings and the differential of the Daemi–Scaduto S-complex of singular instanton homology for the torus knots $T(3, n)$, together with the corresponding facts for two-bridge knots $\mathfrak{b}(p, q)$, whose $(p-1)/2$ irreducible traceless $SU(2)$ characters are all binary dihedral. The irreducible $SU(2)$ characters of $T(3, n)$ form open arcs, each containing at most one traceless character; there are $|\sigma(T(3, n))|/2$ irreducible traceless characters, of which $(\det T(3, n) - 1)/2$ are binary dihedral. For n odd, Daemi and Scaduto’s computation of the irreducible instanton complex confirmed the conjecture of Poudel and Saveliev that the irreducible flat $SO(3)$ connections on the double branched cover $\Sigma(2, 3, n)$ split evenly between gradings 1 and 3. For every torus knot K , the S-complex built on the traceless character variety has rank $1 + |\sigma(K)|$, and Kronheimer–Mrowka’s reduced singular instanton homology $I^\natural(K)$ has rank $\|\Delta_K\|_1$, the sum of the absolute values of the coefficients of the Alexander polynomial (Li–Ye); so the S-complex differential has rank $(1 + |\sigma(K)| - \|\Delta_K\|_1)/2$, a direct consequence recorded here. For $T(3, n)$ this rank is 0 for $n \equiv 1, 2 \pmod{6}$ and 1 for $n \equiv 4, 5 \pmod{6}$, and it equals Daemi and Scaduto’s Frøyshov-type invariant $h(T(3, n))$. Daemi and Scaduto proved the cases $n \equiv 1, 2 \pmod{6}$ and $n = 4, 5$, and Hedden–Herald–Kirk computed the pattern for $n \leq 38$; the formula gives all n . We also prove $\text{rank } I^\natural(T(3, n)) = \|\Delta\|_1$ through Khovanov homology, using Schütz’s decomposition of the Bar–Natan complex of three-braids, and we compare the rank-one differential of $8_{19} = T(3, 4)$ with the single pillowcase bigon found by Hedden–Herald–Kirk.

1. INTRODUCTION

1.1. The Atiyah–Floer conjecture and its knot version. Let Y be a closed oriented 3-manifold with a Heegaard splitting $Y = Y_0 \cup_\Sigma Y_1$. Two Floer theories are associated with this picture. On the gauge-theoretic side is Floer’s instanton homology $HF^{\text{inst}}(Y)$, generated by flat $SU(2)$ connections and with a differential counting anti-self-dual instantons on $Y \times \mathbb{R}$. On the symplectic side, the moduli space $M(\Sigma)$ of flat $SU(2)$ connections on Σ is symplectic, the handlebodies determine Lagrangians $L_0, L_1 \subset M(\Sigma)$, and one forms the Lagrangian Floer homology $HF^{\text{symp}}(L_0, L_1)$. The Atiyah–Floer conjecture [Ati88] asserts that these are isomorphic. Dostoglou and Salamon proved the analogous statement for mapping tori [DS94]. For Heegaard splittings, Salamon and Wehrheim developed instanton Floer theory with Lagrangian boundary conditions [SW08], and Daemi, Fukaya and Lipyanskiy establish a version of the conjecture for admissible $SO(3)$ -bundles [DFL21]; see Auroux’s survey [Aur26] for the place of the conjecture in Lagrangian Floer theory. The original statement for the trivial bundle over an integral homology sphere remains open.

There is a knot-theoretic analogue in which the pillowcase plays the role of $M(\Sigma)$. Let $K \subset S^3$ be a knot, and cut S^3 along a Conway sphere meeting K in four points into two 2-string tangles. A representation $\rho: \pi_1(S^3 \setminus K) \rightarrow SU(2)$ is *traceless* if every meridian maps to a traceless element

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(conjugate to $\text{diag}(i, -i)$); this is the boundary condition of Kronheimer–Mrowka’s singular instanton theory [KM11a, KM11b]. Restricting traceless flat connections of each tangle to the four-punctured sphere and applying holonomy perturbations, one obtains immersed curves L_0, L_1 in the *pillowcase*

$$\mathcal{P} = (\mathbb{R}/2\pi\mathbb{Z})^2 / \{(\gamma, \theta) \sim (-\gamma, -\theta)\},$$

the traceless $\text{SU}(2)$ character variety of the four-punctured sphere: a two-sphere with four $\mathbb{Z}/2$ orbifold points [Lin92, HHK14]. Hedden, Herald and Kirk conjectured that for suitable decompositions, in which one tangle is trivial and carries the earring (the small linking circle that Kronheimer and Mrowka add to the knot to define the reduced theory; Section 3.2), the Lagrangian Floer homology of (L_0, L_1) is isomorphic to Kronheimer–Mrowka’s reduced singular instanton homology $I^\natural(K)$ as a relatively $\mathbb{Z}/4$ -graded group [HHK18, Introduction and Conj. 6.5]. Cazassus, Herald, Kirk and Kotelskiy proposed that an isomorphism with $I^\natural(K)$ holds for arbitrary two-tangle decompositions once L_0 and L_1 carry suitable bounding cochains [CHKK22, Conj. D]. (A bounding cochain on an immersed curve L is an element b of the self-Floer complex $CF(L, L)$ that satisfies the Maurer–Cartan equation $\sum_k \mathfrak{m}_k(b, \dots, b) = 0$; it deforms the Floer differential by counts of higher polygons [Aur26, CHKK22]. Cazassus–Herald–Kirk–Kotelskiy propose figure-eight bubbling, a degeneration of quilted strips that can occur when Lagrangian correspondences are composed, as the mechanism that produces such cochains [CHKK22, §10.4].)

1.2. What this note contains. This note collects, for two families of knots, the data on which the pillowcase approach rests: the generators (the traceless characters), their $\mathbb{Z}/4$ gradings, and the differential of the Daemi–Scaduto S-complex. Most of the individual statements are known, and each is attributed where it is stated. What the note adds is a uniform treatment of the family $T(3, n)$ in all four residue classes of n modulo 6 (Sections 4 and 5); the rank formula of Corollary 1.4, which follows at once from work of Li–Ye and Daemi–Scaduto; and a proof through Khovanov homology, using Schütz’s computation for three-braids, that $\text{rank } I^\natural(T(3, n)) = \|\Delta_{T(3, n)}\|_1$ (Lemma 5.2). In particular, for $n \equiv 4, 5 \pmod{6}$ and $n \geq 10$ the rank-one statement of Theorem 1.3(ii) is deduced here from Corollary 1.4 and the recursion for h in [DS24a, Prop. 6.11].

Traceless character varieties of two-bridge knots also enter the work of Dai–Mallick–Taniguchi, who use them to compute, in part, symmetry actions on singular instanton complexes [DMT26].

1.3. Results. Section 2 recalls the pillowcase and how traceless characters give generators.

Theorem 1.1 (Lewallen; Hedden–Herald–Kirk; Nagasato–Yamaguchi). *Let $K = \mathfrak{b}(p, q)$ be the two-bridge knot of fraction p/q , normalized so that q is odd; then $p = \det K$. Every irreducible traceless $\text{SU}(2)$ representation of $\pi_1(S^3 \setminus K)$ is binary dihedral. There are exactly $(p-1)/2$ of them up to conjugacy: writing ω for the angle between the images of the two meridian generators, they occur at $\cos \omega = \cos(2\pi k/p)$ for $k = 1, \dots, (p-1)/2$, independently of q . Equivalently, the traceless Riley polynomial (Riley’s polynomial at meridian eigenvalue $s = i$, made monic; Section 3) in $u = 2 \cos \omega - 2$ is*

$$\phi_p(u) = \prod_{k=1}^{(p-1)/2} \left(u + 4 \sin^2 \frac{\pi k}{p} \right),$$

of degree $(p-1)/2$, with integer coefficients, $\phi_p(0) = p = \det K$, and sum of roots $-p$.

The binary-dihedral statement and the count are due to Lewallen [Lew09, Prop. 2.3, Thm. 2.4 and the remark after Thm. 2.5]; Hedden–Herald–Kirk give another argument [HHK14, Thm. 10.2], and Nagasato–Yamaguchi prove the statement for trace-free $\text{SL}_2(\mathbb{C})$ characters, with the same factorization of the traceless Riley polynomial [NY12, Thms. 7.5 and 7.6]. The proof in Section 3

is Lewallen’s argument [Lew09, proof of Prop. 2.3] written in quaternions; we include it because it locates the characters explicitly, which is what Section 3.2 and Figure 1 use.

Theorem 1.2 (Klassen; Hedden–Herald–Kirk). *Let $n \geq 2$ and $\gcd(3, n) = 1$. The irreducible traceless $\mathrm{SU}(2)$ characters of $T(3, n)$ are in bijection with the traceless arcs of Section 4 (the arcs of irreducible characters that contain a traceless point), and their number is*

$$N(3, n) = \lfloor 5n/6 \rfloor - \lfloor n/6 \rfloor = -\sigma(T(3, n))/2.$$

Exactly $(\det -1)/2$ of them are binary dihedral: one if n is even ($\det T(3, n) = 3$) and none if n is odd ($\det T(3, n) = 1$).

The count $-\sigma/2$ of irreducible traceless characters of a torus knot is in [HHK14, §5.1] and [DS24b, §9.5], and every knot has exactly $(\det -1)/2$ irreducible binary-dihedral characters [Kla91, Thm. 10]. Section 4 derives the closed form by counting residues in an interval and exhibits the dihedral character for even n .

1.4. Gradings and the S-complex. *Conventions.* $T(3, n)$ denotes the positive torus knot, and signatures are normalized so that positive torus knots have $\sigma < 0$; for example $\sigma(T(2, 3)) = -2$ and $\sigma(T(3, 4)) = -6$ [DS24b, (9.38)]. Chain complexes are homologically graded, with differentials of degree -1 , and $\mathbb{Z}/4$ grading indices are read modulo 4. A $\mathbb{Z}/4$ -graded group whose absolute grading has been fixed is recorded by its rank vector (r_0, r_1, r_2, r_3) , the ranks over $\mathbb{Q}$ in gradings $0, 1, 2, 3$; $\mathbb{Z}_{(k)}^r$ denotes $\mathbb{Z}^r$ in grading k . Three normalizations of the $\mathbb{Z}/4$ instanton grading occur. Daemi–Scaduto’s grading gr_{DS} puts the reducible generator of the S-complex (the summand $\mathbb{Z}_{(0)}$ below, given by the abelian traceless character) in degree 0 [DS24b, (2.18), Def. 3.21]. The grading of $I^\natural(K)$ differs from it by $\sigma(K) \bmod 4$ [DS24b, Thm. 8.9]. For n odd, Poudel and Saveliev grade each irreducible flat $\mathrm{SO}(3)$ connection β on $\Sigma(2, 3, n)$ by $\mu(\beta) \in \mathbb{Z}/4$, defined through an equivariant index on the double branched cover [PS17, Prop. 6.5]; Proposition 5.3 relates μ to gr_{DS} . Rank vectors are given in Daemi–Scaduto’s normalization unless the shift is stated, and a graded group is identified with its rank vector when only ranks matter. (Section 5 also uses the Floer grading of $\Sigma(2, 3, n)$, and Section 6 the pillowcase grading.)

Let K be a torus knot. Its traceless character variety is finite and nondegenerate (each point is a nondegenerate critical point of the unperturbed Chern–Simons functional) [DS24b, §9.5]. Let $C_*(K)$ be Daemi–Scaduto’s irreducible complex, freely generated by the irreducible traceless characters, with differential d . Their S-complex [DS24b, Def. 3.21] is

$$\tilde{C}_*(K) = C_*(K) \oplus C_{*-1}(K) \oplus \mathbb{Z}_{(0)}, \quad \tilde{d} = \begin{pmatrix} d & 0 & 0 \\ v & -d & \delta_2 \\ \delta_1 & 0 & 0 \end{pmatrix},$$

with $v: C_* \rightarrow C_{*-2}$, $\delta_1: C_1 \rightarrow \mathbb{Z}$ and $\delta_2: \mathbb{Z} \rightarrow C_2$. Daemi and Scaduto call it the framed instanton chain complex [DS24b, §3.4]; it is distinct from framed instanton homology $I^\#$. Its homology is isomorphic to $I^\natural(K)$ after the shift by $\sigma(K) \bmod 4$ [DS24b, Thm. 8.9]. The S-complex is defined up to S-chain homotopy equivalence; for the representative above, whose chain group is determined by K , the total rank of $\tilde{d}$ over $\mathbb{Q}$ equals $(\dim \tilde{C} - \dim H_*(\tilde{C}))/2$. Daemi and Scaduto’s invariant $h(K) \in \mathbb{Z}$ is a Frøyshov-type concordance invariant of the local equivalence class of $\tilde{C}(K)$; $h(K) > 0$ if and only if some $\alpha \in C_*$ with $d\alpha = 0$ has $\delta_1(\alpha) \neq 0$ [DS24b, Prop. 4.15].

Theorem 1.3 (Daemi–Scaduto; Hedden–Herald–Kirk for $n \leq 38$). *Let $n \geq 2$ with $\gcd(3, n) = 1$.*

- (i) *For n odd, put $a = -\sigma(T(3, n))/4$. Then $\tilde{C}(T(3, n))$ has rank vector $(1+a, a, a, a)$, and the irreducible flat $\mathrm{SO}(3)$ connections on $\Sigma(2, 3, n)$ split evenly between the Poudel–Saveliev gradings $\mu = 1$ and $\mu = 3$.*

- (ii) *The total rank of $\tilde{d}$ on $\tilde{C}(T(3, n))$ is 0 for $n \equiv 1, 2 \pmod{6}$ and 1 for $n \equiv 4, 5 \pmod{6}$, and in each case it equals $h(T(3, n))$; for $n \equiv 4, 5 \pmod{6}$ the component δ_1 is nonzero.*

Part (i) is the computation of the irreducible torus-knot complex in [DS24a, Cor. 3], which resolved the conjecture of Poudel and Saveliev [PS17, §7.4]; the equivalence of the knot and branched-cover formulations is noted in [DS24b, §9.5], and the case $n \equiv 1 \pmod{6}$ is due to Anvari [Anv19, Ex. 6.2]. In part (ii), the vanishing for $n \equiv 1, 2 \pmod{6}$ is [DS24b, (9.36), Prop. 9.37], and the cases $n = 4, 5$, including $\delta_1 \neq 0$, are [DS24b, §§9.3–9.4]. Hedden–Herald–Kirk computed the whole pattern for $n \leq 38$, from the reduced Khovanov ranks, for the Kronheimer–Mrowka chain complex with their holonomy perturbations, which has the same chain rank and homology [HHK14, §12.4]. For $n \equiv 4, 5 \pmod{6}$ with $n \geq 10$, part (ii) is deduced here: the rank follows from Corollary 1.4, and $\delta_1 \neq 0$ from the recursion for h in [DS24a, Prop. 6.11]. Proposition 5.4 proves part (ii) for all n at once.

Corollary 1.4 (of Li–Ye and Daemi–Scaduto). *For every torus knot K , the total rank of the differential of $\tilde{C}(K)$ is $\frac{1}{2}(1 + |\sigma(K)| - \|\Delta_K\|_1)$, where $\|\Delta_K\|_1$ is the sum of the absolute values of the coefficients of the Alexander polynomial.*

For general torus knots this rank is studied in [Wue26b], which gives a floor-sum formula for it, a sharp lower bound, and the classification of the torus knots for which the differential vanishes.

Section 6 compares the rank-one differential of $8_{19} = T(3, 4)$ with the pillowcase computation of Hedden–Herald–Kirk, and Section 7 states open questions.

1.5. Strategy. The proof of Theorem 1.1 follows Lewallen. Any two traceless elements of $SU(2)$ lie on a common great circle of the sphere of traceless elements, so a representation of a group generated by two meridians is binary dihedral as soon as it is traceless; in the binary-dihedral normal form $\rho(a) = j$, $\rho(b) = e^{i\theta}j$, the two-bridge relation reduces to $e^{i\theta n} = 1$, which places the characters. Riley’s degree bound then identifies the traceless Riley polynomial.

Theorem 1.2 turns on a trace obstruction. Writing $\pi_1(S^3 \setminus T(3, n)) = \langle x, y \mid x^3 = y^n \rangle$, the trace of $\rho(x)$ is ± 1 , so $\rho(x)$ cannot lie in the coset jS^1 of $\text{Pin}(2) = S^1 \cup jS^1$; irreducibility then forces $\rho(y)$ into that coset, where every element is traceless, and that is possible only when n is even. The count reduces to counting residues modulo n in the interval $(n/6, 5n/6)$.

For Theorem 1.3(i), Daemi and Scaduto’s computation [DS24a, Cor. 3] gives $C_* = \mathbb{Z}_{(1)}^a \oplus \mathbb{Z}_{(3)}^a$, and the decomposition $\tilde{C} = C_* \oplus C_{*-1} \oplus \mathbb{Z}_{(0)}$ gives the rank vector. The two traceless characters of $T(3, n)$ lying over a flat connection on $\Sigma(2, 3, n)$ have the same grading, and $\mu = \text{gr}_{DS} + \sigma$, with $\sigma \equiv 0 \pmod{4}$ for n odd; so the knot statement and the cover statement are equivalent. For part (ii) and Corollary 1.4, $\dim \tilde{C} = 1 + |\sigma|$, $\dim H_*(\tilde{C}) = \text{rank } I^\natural = \|\Delta\|_1$, and $\dim H = \dim C - 2 \text{rank } \tilde{d}$. Li–Ye’s theorem gives $\text{rank } I^\natural = \|\Delta\|_1$ for all torus knots; for $T(3, n)$ we also obtain it by squeezing $I^\natural$ between Kronheimer–Mrowka’s Alexander lower bound and the reduced Khovanov upper bound, the latter from Schütz’s chain decomposition.

1.6. Companion papers and computer checks. Three companion papers continue this work. The rank in Corollary 1.4 is studied for all torus knots in [Wue26b], which also determines the ranks over $\mathbb{Q}$ of the graded components of the differential. The pillowcase side of the family $T(3, n)$, $n \geq 4$, is computed in [Wue26c], for the decomposition of [HHK18, §10] given by the integers (r, s) , $3r + ns = 1$, that Hedden, Herald and Kirk use for $T(3, 4)$, $T(3, 5)$ and $T(3, 7)$, and for sufficiently small holonomy perturbations in an explicit closed cone of parameters. For these choices the pillowcase homology has the $\mathbb{Z}/4$ -graded dimensions of $I^\natural$, and, after its grading is shifted by σ and its closed components are suitably graded, the pillowcase complex has the graded generators of $\tilde{C}$ and a differential of the same rank. The pretzel knots $P(-2, 3, q)$, a family that begins with $T(3, 4)$ and $T(3, 5)$, are treated in [Wue26a].

Several statements are also checked numerically, independently of the proofs: the traceless Riley polynomials of Theorem 1.1 for all two-bridge knots with $p \leq 9$, by exact computation of the Riley word; the count of Proposition 4.1 for $n \leq 25$, by enumerating arcs; and the identity (2) and the grading split of Proposition 5.3 for odd $n \leq 43$, by evaluating (4)–(5). The programs are included as ancillary files with this preprint and at <https://github.com/bwuebben/pillowcase-bounding-cochain>, a repository shared with the companion papers [Wue26a, Wue26b, Wue26c]. No proof depends on them.

2. THE PILLOWCASE AND TRACELESS CHARACTERS

We work with $SU(2)$ as the unit quaternions. An element is traceless iff it is a purely imaginary unit quaternion (eigenvalues $\pm i$, squaring to -1); as a rotation it is a rotation by π . The traceless elements form a 2-sphere $S \subset \text{Im } \mathbb{H} \cong \mathbb{R}^3$.

Proposition 2.1 (Classical; cf. [Lin92], [HHK14, Prop. 3.1]). *The traceless $SU(2)$ character variety of the four-punctured sphere is the pillowcase $\mathcal{P}$, with four $\mathbb{Z}/2$ orbifold points at $(\gamma, \theta) \in \{0, \pi\}^2$ and the symplectic form descending from $d\gamma \wedge d\theta$.*

Sketch. Write π_1 of the four-punctured sphere as $\langle X_1, \dots, X_4 \mid X_1 X_2 X_3 X_4 = 1 \rangle$, and write X_i also for its image, an element of S . Since purely imaginary units satisfy $X_i^{-1} = -X_i$, the relation gives $X_1 X_2 = X_4 X_3 =: A$. For pure imaginary units, $X_i X_j = -\langle X_i, X_j \rangle + X_i \times X_j$, so when $A \neq \pm 1$ its axis is well defined; equating the two expressions for A forces the four X_i to be coplanar, orthogonal to $\text{axis}(A)$. Placing that axis along k and writing $X_m = \cos t_m i + \sin t_m j$, the relation becomes $t_2 - t_1 = t_3 - t_4$; gauge-fixing $t_1 = 0$ and setting $\gamma = t_2$, $\theta = t_4$ leaves two parameters, and the residual conjugation by the π -rotation about i acts by $(\gamma, \theta) \mapsto (-\gamma, -\theta)$. The four fixed points are the reducible (collinear) characters. $\square$

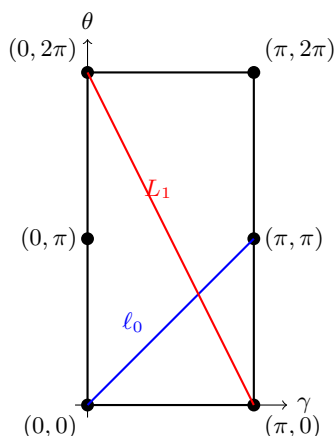


FIGURE 1. The pillowcase $\mathcal{P}$ drawn as the rectangle $[0, \pi] \times [0, 2\pi]$: each vertical edge is folded at its midpoint, $(\gamma, \theta) \sim (\gamma, 2\pi - \theta)$ for $\gamma \in \{0, \pi\}$, and the horizontal edges are identified, so $(0, 2\pi) = (0, 0)$ and $(\pi, 2\pi) = (\pi, 0)$. The four orbifold points are $(0, 0)$, $(\pi, 0)$, $(0, \pi)$ and (π, π) . For the trefoil $\mathfrak{b}(3, 1)$ the trivial tangle gives the diagonal arc $\ell_0 = \{\gamma = \theta\}$ and the rational tangle the straight arc $L_1 = \{\theta = 2\pi - 2\gamma\}$; they meet at the corner $(0, 0)$ and at the interior point $(2\pi/3, 2\pi/3)$ (Section 3.2). The earring correspondence replaces ℓ_0 by a figure-eight curve L_0 near it, whose single self-crossing lies near the midpoint of ℓ_0 and whose two lobes encircle its end corners (Remark 3.2).

For a knot K with a two-tangle decomposition, a traceless representation of $\pi_1(S^3 \setminus K)$ restricts to a compatible pair of tangle representations and so determines a point of $L_0 \cap L_1$; for two-bridge knots this gives a bijection between the intersection points and the traceless characters of the knot group [HHK14, §10], which after perturbation give the generators of the singular instanton chain complex [KM11a, KM11b]. A two-bridge knot group is generated by two meridians a, b , and an irreducible traceless representation is determined up to conjugacy by the single invariant $\cos \omega = \langle \rho(a), \rho(b) \rangle$, the cosine of the angle between $\rho(a), \rho(b) \in S$.

3. TWO-BRIDGE KNOTS

3.1. Traceless characters are binary dihedral. The group of $\mathfrak{b}(p, q)$, with q odd, is $\langle a, b \mid wa = bw \rangle$ with $w = a^{\varepsilon_1} b^{\varepsilon_2} a^{\varepsilon_3} \cdots b^{\varepsilon_{p-1}}$ and $\varepsilon_i = (-1)^{\lfloor iq/p \rfloor}$, where a, b are meridians [Ril84]. Riley's parametrization [Ril84] of the nonabelian $\mathrm{SL}_2(\mathbb{C})$ representations with meridian eigenvalue s is, after a diagonal conjugation,

$$\rho(a) = \begin{pmatrix} s & 1 \\ 0 & s^{-1} \end{pmatrix}, \quad \rho(b) = \begin{pmatrix} s & 0 \\ -u & s^{-1} \end{pmatrix},$$

cut out by a polynomial $\Phi_{p,q}(s, u)$ of degree $(p-1)/2$ in u ; the traceless locus is $s = i$. On that locus $\mathrm{tr} \rho(ab) = s^2 + s^{-2} - u = -2 - u$, and for traceless $\rho(a), \rho(b) \in S$ one has $\mathrm{tr} \rho(ab) = 2 \mathrm{Re}(\rho(a)\rho(b)) = -2 \cos \omega$. Hence Riley's parameter is $u = -2 - \mathrm{tr} \rho(ab) = 2 \cos \omega - 2$. Write ϕ_p for the monic normalization of $\Phi_{p,q}(i, \cdot)$, the traceless Riley polynomial; Theorem 1.1 shows it is independent of q .

Proof of Theorem 1.1. Let ρ be traceless. The unit imaginary quaternions $\rho(a), \rho(b)$ lie in a plane through the origin, so after conjugation $\rho(a) = j$ and $\rho(b) = e^{i\eta} j = \cos \eta j + \sin \eta k$. Both lie in the coset jS^1 of $\mathrm{Pin}(2) = S^1 \cup jS^1$, where $S^1 = \{e^{it}\}$, so ρ is binary dihedral. Since $a^{\pm 1} \mapsto \pm j$, $b^{\pm 1} \mapsto \pm e^{i\eta} j$ and $je^{i\eta} j = -e^{-i\eta}$, each of the $(p-1)/2$ consecutive pairs $a^{\varepsilon} b^{\varepsilon'}$ in w maps to $\pm e^{-i\eta}$, so $\rho(w) = \pm e^{-i(p-1)\eta/2}$ and

$$\rho(w)\rho(a)\rho(w)^{-1} = e^{-i(p-1)\eta} j.$$

The relation $wa = bw$ therefore holds iff $e^{-i(p-1)\eta} = e^{i\eta}$, that is, iff $\eta = 2\pi k/p$ for an integer k . Here $\cos \omega = \langle j, e^{i\eta} j \rangle = \cos \eta$; $k = 0$ gives the abelian representation, and conjugation by j sends η to $-\eta$. The irreducible traceless characters are therefore exactly those with $k = 1, \dots, (p-1)/2$, with distinct values $\cos \omega = \cos(2\pi k/p)$ that depend only on p .

Each of them is a nonabelian representation with $s = i$, so $\Phi_{p,q}(i, u_k) = 0$ at $u_k = 2 \cos(2\pi k/p) - 2 = -4 \sin^2(\pi k/p)$. Since $\Phi_{p,q}(i, \cdot)$ is a nonzero polynomial of degree at most $(p-1)/2$ [Ril84], [NY12, Thm. 7.6], it has no other roots, and $\phi_p(u) = \prod_k (u - u_k)$, as in [NY12, Thm. 7.6]. Its constant term is $\prod_k 4 \sin^2(\pi k/p) = p$ and the sum of its roots is $\sum_k (2 \cos(2\pi k/p) - 2) = -1 - (p-1) = -p$. The Galois group of the p th cyclotomic field permutes the roots u_k , so ϕ_p has rational coefficients; they are algebraic integers because the roots are, and hence lie in $\mathbb{Z}$. $\square$

Remark 3.1. Small cases: $\phi_3 = u + 3$, $\phi_5 = u^2 + 5u + 5$, $\phi_7 = u^3 + 7u^2 + 14u + 7$, $\phi_9 = u^4 + 9u^3 + 27u^2 + 30u + 9$. For p prime, $\phi_p(x-2)$ is the minimal polynomial of $2 \cos(2\pi/p)$, irreducible of degree $(p-1)/2$. The two two-bridge knots of determinant 9, the torus knot $9_1 = \mathfrak{b}(9, 1)$ and the hyperbolic knot $6_1 = \mathfrak{b}(9, 2)$, have the same traceless Riley polynomial ϕ_9 .

3.2. The pillowcase picture and the earring. The traceless character variety of a rational tangle is a straight arc in $\mathcal{P}$ joining two corners, of rational slope determined by the tangle [HHK14, Thm. 3]. Under the branched double cover $(\mathbb{R}/2\pi\mathbb{Z})^2 \rightarrow \mathcal{P}$ these arcs lift to straight lines [Lin92, HHK14, CHKK22]. For $\mathfrak{b}(p, q)$ the two lifted lines have directions $(1, 1)$ and $(q, q-p)$ in the

coordinates (γ, θ) [HHK14, Thm. 3] and meet in $|\det(\frac{1}{q} \frac{1}{q-p})| = p$ points; correspondingly, $\Sigma_2(K) = L(p, q)$ is the union of two solid tori, the double branched covers of the two rational tangles.

The naive count on $\mathcal{P}$ itself is $(p+1)/2$: the $(p-1)/2$ interior dihedral characters of Theorem 1.1 together with the single abelian character at a corner. The passage to $\det = p$ comes from the *earring*: the reduced theory $I^\natural$ is defined by adding a small earring to the knot [KM11b, §4.3], and in the pillowcase picture this replaces the curve of the tangle that carries the earring by its image under the earring correspondence E [CHKK22, HK24]. For the trivial tangle the image of the diagonal arc ℓ_0 is a figure-eight curve $L_0 = E(\ell_0)$ close to it [HHK14, Thm. 1]. Each of the $(p-1)/2$ interior intersection points of the straight arc L_1 with ℓ_0 gives two intersection points with L_0 , and the corner gives one, for p in all. Since L_1 is straight and L_0 is close to ℓ_0 , there are no immersed bigons. Hedden–Herald–Kirk show that for every two-bridge knot the differentials vanish, the pillowcase homology has rank $\det K$, and the relative $\mathbb{Z}/4$ gradings agree with those of $I^\natural$ [HHK18, §7], with the grading identification outlined in [HHK18, §6.1]. Theorem 1.1 adds the identification of the interior generators with the binary-dihedral characters.

Remark 3.2 (The earring correspondence). Herald and Kirk prove, at the level of regular homotopy and in the limit of small perturbation, that E replaces an immersed arc joining two corners (satisfying a mild genericity condition, ‘good’ in the sense of [HK24]) by a figure-eight curve near it, an immersed circle with a single transverse self-crossing near the midpoint of the arc whose two lobes encircle the end corners, and replaces an immersed circle avoiding the corners by two parallel copies [HK24, Thm. B, §6.4.2, Thms. 6.8 and 6.15]. Thus $E \circ L$ is immersed even when L is embedded. For a two-bridge knot the only self-crossing at which a bounding cochain could sit is the one near the midpoint of the arc, and Smith shows that the cochain associated to a rational tangle with an earring vanishes if [CHKK22, Conj. D] holds [Smi24, Lemma 5.3]. Off the two-bridge family there is evidence that some decompositions require nonzero bounding cochains: for the $(4, 5)$ -torus knot, Cazassus–Herald–Kirk–Kotelskiy show that with zero cochains the Floer homology depends on which tangle of the decomposition of [HHK18, §11.8] carries the earring, and argue informally that a nonzero cochain restores agreement [CHKK22, §10.3]; Smith proves that if their Conjecture D holds, then some tangle without earring must carry a nonzero cochain, exhibited in a decomposition of $P(-2, 3, 5)$ [Smi24, Thm. 5.9]. For a second decomposition of $T(3, 4)$ in the family of [HHK18, §10], a numerical computation gives pillowcase homology of rank 7 with zero cochains, against rank $I^\natural(T(3, 4)) = 5$, and rank 5 with a nonzero cochain at a self-intersection of the arc [Wue26c, §7]. For Smith’s decomposition of $P(-2, 3, 7)$, the pairing of the earring curve with a piecewise-linear model of Smith’s curve, computed with zero cochain in an algebraic model of the wrapped Fukaya category, has dimension 7, against rank $I^\natural(P(-2, 3, 7)) = 9$, and Maurer–Cartan deformations of the model raise it to 9 [Wue26a].

4. THE $(3, n)$ -TORUS KNOTS

4.1. The representation arcs. Throughout this section $n \geq 2$ and $\gcd(3, n) = 1$. Write $\pi_1(S^3 \setminus T(3, n)) = \langle x, y \mid x^3 = y^n \rangle$. The element $z = x^3 = y^n$ is central, so any irreducible $\mathrm{SU}(2)$ representation sends $z \mapsto \pm 1$ and

$$\rho(x) = e^{\vartheta_1 \hat{u}}, \quad \rho(y) = e^{\vartheta_2 \hat{v}}, \quad \vartheta_1 = \frac{\pi \ell_1}{3}, \quad \vartheta_2 = \frac{\pi \ell_2}{n},$$

with $\hat{u}, \hat{v}$ unit imaginary quaternions, $\ell_1 \in \{1, 2\}$, $0 < \ell_2 < n$, and $\ell_1 \equiv \ell_2 \pmod{2}$, since $\rho(x)^3 = \rho(y)^n = \rho(z)$. The angle $\psi = \angle(\hat{u}, \hat{v}) \in (0, \pi)$ is the only remaining modulus; each such pair (ℓ_1, ℓ_2) is a one-parameter arc of irreducible characters [Kla91, Thm. 1]. Call an arc a *traceless arc* if it contains a traceless character.

4.2. The traceless locus. Fix integers c, f with $cn + 3f = 1$. The meridian of $T(3, n)$ is $x^c y^f$ [HHK14, §5.1]; it does not depend on the choice of (c, f) , since $x^{c+3t} y^{f-nt} = x^c z^t y^f z^{-t} = x^c y^f$. A direct quaternion computation gives

$$\frac{1}{2} \operatorname{tr} \rho(x^c y^f) = \operatorname{Re} \rho(x^c y^f) = \cos(c\vartheta_1) \cos(f\vartheta_2) - \sin(c\vartheta_1) \sin(f\vartheta_2) \cos \psi,$$

which is affine in $\cos \psi$. Hence each arc contains exactly one traceless character, at

$$\cos \psi = \cot(c\vartheta_1) \cot(f\vartheta_2), \quad (1)$$

provided the right side lies in $(-1, 1)$; otherwise the arc carries none, and the boundary case never occurs, as the proof below shows. By construction $\rho(x)^3 = \rho(y)^n = (-1)^{\ell_1}$.

Proposition 4.1 (Count). *There are*

$$N(3, n) = \lfloor 5n/6 \rfloor - \lfloor n/6 \rfloor = \begin{cases} 2(n-1)/3, & n \equiv 1 \pmod{6}, \\ (2n-1)/3, & n \equiv 2 \pmod{6}, \\ (2n+1)/3, & n \equiv 4 \pmod{6}, \\ 2(n+1)/3, & n \equiv 5 \pmod{6} \end{cases}$$

irreducible traceless $\mathrm{SU}(2)$ characters of $T(3, n)$, and in addition exactly one abelian traceless character. Moreover $N(3, n) = -\sigma(T(3, n))/2$ [HHK14, §5.1], [DS24b, §9.5], and for n odd half of the irreducible traceless characters have $\rho(z) = 1$ and half have $\rho(z) = -1$.

Proof. From $cn + 3f = 1$ we get $\gcd(c, 3) = \gcd(f, n) = 1$. Since $\ell_1 \in \{1, 2\}$ and $3 \nmid c\ell_1$, the first cotangent in (1) always has $|\cot(c\vartheta_1)| = \cot(\pi/3) = 1/\sqrt{3}$; so an arc is a traceless arc iff $|\cot(f\vartheta_2)| < \sqrt{3}$, iff $\pi f\ell_2/n$ lies in $(\pi/6, 5\pi/6)$ modulo π , iff the residue $\lambda = f\ell_2 \bmod n$ lies in the open interval $(n/6, 5n/6)$. Equality never occurs: it would force $6 \mid n$, impossible with $\gcd(3, n) = 1$, so the boundary case $\cos \psi = \pm 1$ does not arise. The parity condition assigns to each $\ell_2 \in \{1, \dots, n-1\}$ exactly one ℓ_1 ($\ell_1 = 1$ for ℓ_2 odd, $\ell_1 = 2$ for ℓ_2 even), and $\ell_2 \mapsto \lambda = f\ell_2 \bmod n$ is a bijection of $\{1, \dots, n-1\}$; hence

$$N(3, n) = \#\{\lambda \in \mathbb{Z} : n/6 < \lambda < 5n/6\} = \lfloor 5n/6 \rfloor - \lfloor n/6 \rfloor,$$

which evaluates, in the four residues $n \equiv 1, 2, 4, 5 \pmod{6}$, to the stated cases. Distinct arcs give distinct characters, since $\operatorname{tr} \rho(y) = 2 \cos(\pi\ell_2/n)$ separates them, and the abelian traceless character (every meridian to a fixed traceless element) is unique up to conjugacy. Finally, for n odd the map $(\ell_1, \ell_2) \mapsto (3 - \ell_1, n - \ell_2)$ respects the parity condition and exchanges $\rho(z) = -1$ with $\rho(z) = 1$; it preserves the condition $\lambda \in (n/6, 5n/6)$, since $\lambda \mapsto n - \lambda$, and it preserves ψ in (1), since both cotangents change sign. Being fixed-point-free, it splits the $N(3, n)$ traceless characters evenly between the two signs of $\rho(z)$. $\square$

4.3. The dihedral dichotomy.

Proof of Theorem 1.2. The bijection and the count are Proposition 4.1. Conjugate a binary-dihedral image into $\operatorname{Pin}(2) = S^1 \cup jS^1$. Since $\operatorname{tr} \rho(x) = 2 \cos(\pi\ell_1/3) = \pm 1$, the element $\rho(x)$ cannot lie in the coset jS^1 , whose elements are traceless; it therefore lies in S^1 . Irreducibility forces $\rho(y)$ to lie in jS^1 . Thus $\vartheta_2 = \pi/2$, equivalently $\ell_2 = n/2$, so n must be even.

Conversely, suppose n is even and take $\ell_2 = n/2$. The parity condition determines a unique $\ell_1 \in \{1, 2\}$. In the relation $cn + 3f = 1$ the integer f is odd, and hence $\cot(f\vartheta_2) = \cot(f\pi/2) = 0$. Equation (1) gives $\cos \psi = 0$. Relative to the circle S^1 containing $\rho(x)$, this places $\rho(y)$ in jS^1 ; the resulting traceless character is irreducible and binary dihedral. Hence there is exactly one such character for even n and none for odd n . This equals $(\det T(3, n) - 1)/2$, since $\det T(3, n) = |\Delta_{T(3, n)}(-1)|$ is 3 for even n and 1 for odd n (Lemma 5.2 below), in agreement with [Kla91, Thm. 10]. $\square$

Theorem 1.2 contrasts with Theorem 1.1: the irreducible traceless characters of a two-bridge knot are all binary dihedral, those of $T(3, n)$ with n odd none. In the pillowcase pictures of [HHK18, §§11.6–11.7] for $T(3, 4)$ and $T(3, 5)$ the non-dihedral characters lie on straight arcs of a singular tangle character variety, and it is the holonomy perturbation resolving that variety which produces non-straight curves and, with them, bigons (Section 6).

5. GRADINGS AND THE S-COMPLEX DIFFERENTIAL

The double branched cover of $T(3, n)$ is the Brieskorn manifold $\Sigma(2, 3, n)$ [PS17, §7.4]. For odd n it is an integral homology sphere; its irreducible flat $\mathrm{SO}(3)$ connections have rotation numbers $(m_1, m_2, m_3) = (1, m_2, m_3)$ indexing the three singular fibers, with m_2, m_3 even, $0 < m_2 < 3$, $0 < m_3 < n$. Since 2 is the only even value in $(0, 3)$, necessarily $m_2 = 2$, and m_3 ranges over the even integers in $(0, n)$ meeting the spherical-triangle inequalities on the angles $(\pi/2, 2\pi/3, \pi m_3/n)$. There are $a = -\sigma(T(3, n))/4$ such connections, all nondegenerate [FS90], [PS17, §7.4], and all invariant under the covering involution [PS17, §7.4], [DS24b, §9.5]. A traceless character α of $T(3, n)$ lies over a flat connection β on $\Sigma(2, 3, n)$ if the adjoint of α pulls back to β . Over each irreducible β lie exactly two irreducible traceless characters, exchanged by the flip ι , which twists a representation by the sign character sending each meridian to -1 [PS17, Prop. 4.1(c)], [DS24b, §§2.3, 9.1]. Consequently

$$N(3, n) = 2a \quad (n \text{ odd}). \quad (2)$$

5.1. The grading formula. The proof of Proposition 5.3 does not use the formulas of this subsection. In Poudel and Saveliev’s description of $I^\natural$ through equivariant gauge theory on the double branched cover, each irreducible β contributes a pair of circles of critical points, exchanged by the flip [PS17, Prop. 4.1(c)]. Both circles are graded by $\mu(\beta)$ [PS17, Prop. 6.5], each contributes two generators, in gradings $\mu(\beta)$ and $\mu(\beta) + 1$, and the trivial connection Θ contributes one generator, in grading $\sigma \bmod 4$. Anvari evaluates μ explicitly [Anv19, §6]:

$$\mu(\beta) = \frac{1}{2} \mathrm{gr}_{FS}(\beta) + \frac{1}{4} (1 - \rho_{\mathrm{Ad} \beta}) \pmod{4}, \quad (3)$$

where $\mathrm{gr}_{FS}(\beta)$ is an integer whose reduction mod 8 is the Floer grading of β in the instanton homology of $\Sigma(2, 3, n)$ [FS90], and $\rho_{\mathrm{Ad} \beta}$ is the equivariant ρ -invariant. In Anvari’s normalization [Anv19, (3), Thm. B],

$$\mathrm{gr}_{FS}(\beta) = \frac{e^2}{3n} + \frac{2}{3} \sum_{k=1}^2 \cot \frac{\pi k}{3} \cot \frac{\pi b_2 k}{3} \sin^2 \frac{\pi k m_2}{3} + \frac{2}{n} \sum_{k=1}^{n-1} \cot \frac{\pi k}{n} \cot \frac{\pi b_3 k}{n} \sin^2 \frac{\pi k m_3}{n}, \quad (4)$$

$$\rho_{\mathrm{Ad} \beta} = 1 - \frac{2}{3n} - \frac{4}{3} \sum_{k=1}^2 \cot \frac{\pi k}{3} \tan \frac{\pi b_2 k}{3} \cos^2 \frac{\pi k m_2}{3} - \frac{4}{n} \sum_{k=1}^{n-1} \cot \frac{\pi k}{n} \tan \frac{\pi b_3 k}{n} \cos^2 \frac{\pi k m_3}{n}, \quad (5)$$

with $e = 3n m_1 + 2n m_2 + 6 m_3$, where b_1, b_2, b_3 are Seifert invariants of $\Sigma(2, 3, n)$, integers with $3n b_1 + 2n b_2 + 6 b_3 = 1$ and b_2, b_3 even; since (4)–(5) depend on b_2 only modulo 3 and on b_3 only modulo n , one may take $b_2 \equiv (2n)^{-1} \pmod{3}$ and $b_3 \equiv 6^{-1} \pmod{n}$. These formulas apply only for n odd, where $\Sigma(2, 3, n)$ is a homology sphere; the rank argument of Section 5.4 does not use the parity of n . By Proposition 5.3, for each odd n the split of Theorem 1.3(i) is an identity among the trigonometric sums (4)–(5): the values (3) are 1 and 3 equally often. Daemi and Scaduto note this consequence of their computation for Anvari’s formulas [DS24a]. The ρ -invariant term is needed: on $\Sigma(2, 3, 11)$ the four connections have Floer gradings 1, 7, 5, 3 (mod 8) and ρ -invariants $-1, 3, -1, 3$, so with a constant ρ -term the values (3) would be four distinct residues.

Example 5.1. For $\Sigma(2, 3, 7)$ the two connections $(1, 2, 2)$ and $(1, 2, 4)$ have $\mathrm{gr}_{FS} \equiv 7$ and 3 (mod 8) and $\rho_{\mathrm{Ad} \beta} = 3$ for both, hence $\mu = 3$ and $\mu = 1$, as in [Anv19, Ex. 6.1].

5.2. The homology ranks.

Lemma 5.2 (Li–Ye; a second proof via Khovanov homology). *For every $n \geq 1$ coprime to 3,*

$$\operatorname{rank}_{\mathbb{Q}} I^{\natural}(T(3, n); \mathbb{Q}) = \|\Delta_{T(3, n)}\|_1 = \begin{cases} 4k + 1, & n = 3k + 1, \\ 4k + 3, & n = 3k + 2, \end{cases}$$

and $|\Delta_{T(3, n)}(-1)|$ is 1 for n odd and 3 for n even.

The equality $\operatorname{rank} I^{\natural} = \|\Delta\|_1$ is the case of $T(3, n)$ in Li–Ye’s computation of the sutured instanton knot homology KHI of Kronheimer and Mrowka for torus knots [LY22, Cor. 1.8], combined with $I^{\natural}(K; \mathbb{Q}) \cong KHI(K; \mathbb{Q})$ [KM11b, Prop. 1.4]; Hedden–Herald–Kirk checked it for $n \leq 38$ [HHK14, §12.4], and the cases $n = 4, 5$ are in [DS24b, §§9.3–9.4]. The following proof replaces Li–Ye’s computation by Schütz’s computation of reduced Khovanov homology, combined with the Kronheimer–Mrowka bounds.

Proof. Schütz works over the Bar–Natan ring $\mathbb{Z}[X, Y]/(X^2 - XY)$ (he writes h for Y ; this h is unrelated to the invariant $h(K)$) and writes A for its q -shifted rank-one free module and $A(j)$ for the two-term complex with differential $(2X - Y)^j$ [Sch25, Def. 3.2]; the reduced Khovanov complex is obtained by tensoring the Bar–Natan complex with $\mathbb{Z}$, where X and Y act by zero [Sch25, §3]. His chain decompositions give, with grading shifts suppressed,

$$\begin{aligned} C_{BN}(T(3, 3k + 1)) &\simeq A \oplus A(1)^k \oplus A(2)^k, \\ C_{BN}(T(3, 3k + 2)) &\simeq A \oplus A(1)^{k+1} \oplus A(2)^k \end{aligned}$$

[Sch25, Thm. 5.3(2),(4)]. After setting $X = Y = 0$, A contributes one generator and each $A(j)$ a complex on two generators with zero differential, so the reduced complex is chain homotopy equivalent to a complex with $1 + 2k + 2k = 4k + 1$, respectively $1 + 2(k + 1) + 2k = 4k + 3$, generators and zero differential. Consequently

$$\operatorname{rank}_{\mathbb{Q}} \widetilde{Kh}(T(3, n); \mathbb{Q}) = \begin{cases} 4k + 1, & n = 3k + 1, \\ 4k + 3, & n = 3k + 2. \end{cases} \quad (6)$$

Up to multiplication by a monomial, the Alexander polynomial is

$$\Delta_{T(3, n)}(t) = \frac{1 + t^n + t^{2n}}{1 + t + t^2},$$

so $\Delta_{T(3, n)}(-1) = 2 + (-1)^n$. Below degree n its coefficients repeat $1, -1, 0$, and the quotient is symmetric of degree $2n - 2$. The central coefficient, in degree $n - 1$, has absolute value one. If $n = 3k + 1$, the coefficients in degrees below $n - 1$ include $2k$ nonzero ones, each ± 1 ; if $n = 3k + 2$, they include $2k + 1$. By the symmetry of Δ , $\|\Delta_{T(3, n)}\|_1 = 2 \cdot 2k + 1 = 4k + 1$, respectively $2(2k + 1) + 1 = 4k + 3$, the right side of (6). The group $KHI(K)$ is Alexander-graded with graded Euler characteristic $\pm \Delta_K(t)$ [KM10, Thm. 1.1], so $\operatorname{rank} KHI(K) \geq \|\Delta_K\|_1$ [KM10, Cor. 1.2]. The reduced Khovanov-to-instanton spectral sequence gives $\operatorname{rank} \widetilde{Kh}(K) \geq \operatorname{rank} I^{\natural}(K)$ [KM11b, Prop. 1.2, Cor. 1.3]. Together with $I^{\natural}(K; \mathbb{Q}) \cong KHI(K; \mathbb{Q})$ [KM11b, Prop. 1.4], this yields

$$\|\Delta_{T(3, n)}\|_1 \leq \operatorname{rank}_{\mathbb{Q}} I^{\natural}(T(3, n); \mathbb{Q}) \leq \operatorname{rank}_{\mathbb{Q}} \widetilde{Kh}(T(3, n); \mathbb{Q}) = \|\Delta_{T(3, n)}\|_1,$$

and equality holds throughout. $\square$

5.3. The grading split for n odd.

Proposition 5.3 (Daemi–Scaduto). *Let n be odd and coprime to 3, and set $a = -\sigma(T(3, n))/4 = N(3, n)/2$. Then $\tilde{C}(T(3, n))$ has rank vector $(1 + a, a, a, a)$, and $a/2$ of the irreducible flat $\mathrm{SO}(3)$ connections on $\Sigma(2, 3, n)$ have $\mu = 1$ and $a/2$ have $\mu = 3$.*

Proof. Daemi and Scaduto prove that for every torus knot K the irreducible complex is $C_*(K) \cong \mathbb{Z}_{(1)}^{[-\sigma(K)/4]} \oplus \mathbb{Z}_{(3)}^{[-\sigma(K)/4]}$, with zero differential [DS24a, Cor. 3], [DS24b, §9.5]. For n odd, $-\sigma/4 = a$, so $C_* = \mathbb{Z}_{(1)}^a \oplus \mathbb{Z}_{(3)}^a$, and $\tilde{C} = C_* \oplus C_{*-1} \oplus \mathbb{Z}_{(0)}$ has rank vector $(1 + a, a, a, a)$; this is the rank vector conjectured by Poudel and Saveliev [PS17, §7.4].

Let β be an irreducible flat connection on $\Sigma(2, 3, n)$ and $\alpha, \iota\alpha$ the two characters lying over it. They have the same grading [PS17, Lemma 6.4 and the paragraph following it]; this also follows from the flip symmetry $M(\alpha, \alpha') \rightarrow M(\iota\alpha, \iota\alpha')$ of the moduli spaces [DS24b, §9.1]. Poudel and Saveliev define $\mu(\beta) = \mathrm{ind} D_B^T(\beta, \Theta) + \sigma \pmod{4}$, where $\mathrm{ind} D_B^T$ is the index of the deformation operator on deck-invariant sections over $\mathbb{R} \times \Sigma(2, 3, n)$ [PS17, Prop. 6.5]. This equivariant index equals the orbifold index over $\mathbb{R} \times (S^3, T(3, n))$ [PS17, §3.5, (7)], which is $\mathrm{gr}_{DS}(\alpha)$ [DS24b, (2.11), (2.18)]; both use weighted spaces with decay at the reducible end. Hence

$$\mu(\beta) = \mathrm{gr}_{DS}(\alpha) + \sigma(T(3, n)) \pmod{4},$$

consistent with the degree $\sigma \pmod{4}$ of the isomorphism of [DS24b, Thm. 8.9]; Daemi and Scaduto use the same identification of gradings in [DS24b, §9.4]. For n odd, $\sigma = -2N(3, n) = -4a \equiv 0 \pmod{4}$ by (2), so each β contributes two generators of C_* in degree $\mu(\beta)$. If ν_j denotes the number of β with $\mu(\beta) = j$, the ranks of C_* give $\nu_1 = \nu_3 = a/2$ and $\nu_0 = \nu_2 = 0$. In particular a is even; explicitly $a = (n - 1)/3$ for $n \equiv 1 \pmod{6}$ and $a = (n + 1)/3$ for $n \equiv 5 \pmod{6}$. $\square$

5.4. The S-complex differential.

Proposition 5.4 (Daemi–Scaduto for $n \equiv 1, 2 \pmod{6}$ and $n = 4, 5$). *Let $n \geq 2$ be coprime to 3 and $N = N(3, n)$. The S-complex $\tilde{C}(T(3, n))$ has rank $2N + 1$, and the total rank of its differential $\tilde{d}$ is*

$n \pmod{6}$	N	$\mathrm{rank}_{\mathbb{Q}} I^{\natural}(T(3, n); \mathbb{Q})$	$\mathrm{rank}_{\mathbb{Q}} \tilde{d}$
1	$2(n - 1)/3$	$2N + 1$	0
2	$(2n - 1)/3$	$2N + 1$	0
4	$(2n + 1)/3$	$2N - 1$	1
5	$2(n + 1)/3$	$2N - 1$	1.

In each case $\mathrm{rank} \tilde{d} = h(T(3, n))$, and for $n \equiv 4, 5 \pmod{6}$ the component δ_1 of $\tilde{d}$ is nonzero.

Proof. The irreducible complex has one generator for every irreducible traceless character, and these are nondegenerate [DS24b, §9.5]; so $\tilde{C} = C_* \oplus C_{*-1} \oplus \mathbb{Z}_{(0)}$ has rank $2N + 1$, with N from Proposition 4.1. Lemma 5.2 gives the homology ranks in the table ($n = 6m + 1, 6m + 2, 6m + 4, 6m + 5$ give $\|\Delta\|_1 = 8m + 1, 8m + 3, 8m + 5, 8m + 7$), and the identity $\dim H = \dim C - 2 \mathrm{rank} \tilde{d}$ applied to $\tilde{C}$, whose homology has the rank of $I^{\natural}$ [DS24b, Thm. 8.9], gives the differential ranks.

For the remaining assertions we use the invariant h . Specialized to the torus knots $T(3, n)$, the recursion of Daemi and Scaduto [DS24a, Prop. 6.11] reads

$$h(T(3, n + 3)) + \frac{1}{2}\sigma(T(3, n + 3)) = h(T(3, n)) + \frac{1}{2}\sigma(T(3, n)) - 2.$$

With $\sigma = -2N$ and the values of N above, $N(3, n + 3) - N(3, n)$ is 3 for $n \equiv 1, 2$ and 1 for $n \equiv 4, 5 \pmod{6}$. So h increases by one from $n \equiv 1, 2$ to $n + 3 \equiv 4, 5$, and decreases by one from $n \equiv 4, 5$ to $n + 3 \equiv 1, 2 \pmod{6}$. Starting from the unknot $T(3, 1)$, whose S-complex is $\mathbb{Z}_{(0)}$ with zero differential, so that $h = 0$, and the trefoil $T(3, 2)$, for which $h = 0$ [DS24b, Thm. 1.14(i)],

this gives $h(T(3, n)) = 0$ for $n \equiv 1, 2 \pmod{6}$ and $h(T(3, n)) = 1$ for $n \equiv 4, 5 \pmod{6}$. By [DS24b, Prop. 4.15], $h > 0$ holds iff there is $\alpha \in C_*$ with $d\alpha = 0$ and $\delta_1(\alpha) \neq 0$; since $d = 0$, this gives $\delta_1 \neq 0$. $\square$

Proof of Theorem 1.3. Part (i) is Proposition 5.3; part (ii) is Proposition 5.4. $\square$

Proof of Corollary 1.4. For a torus knot the traceless character variety consists of $|\sigma(K)|/2$ irreducible points and one abelian point, all nondegenerate [DS24b, §9.5], so $\dim \tilde{C}(K) = 1 + |\sigma(K)|$. Its homology has the rank of $I^\natural(K)$ [DS24b, Thm. 8.9], and $\text{rank } I^\natural(K) = \dim KHI(K) = \|\Delta_K\|_1$ by [KM11b, Prop. 1.4] and [LY22, Cor. 1.8] (Li and Ye work over $\mathbb{C}$, which does not change dimensions). The identity $\dim H = \dim C - 2 \text{rank } \tilde{d}$ gives the formula. $\square$

n	2	4	5	7	8	10	11	13	14	16	17
$n \bmod 6$	2	4	5	1	2	4	5	1	2	4	5
$N(3, n)$	1	3	4	4	5	7	8	8	9	11	12
$\dim \tilde{C} = 2N + 1$	3	7	9	9	11	15	17	17	19	23	25
$\text{rank } I^\natural = \ \Delta\ _1$	3	5	7	9	11	13	15	17	19	21	23
$\text{rank } \tilde{d}$	0	1	1	0	0	1	1	0	0	1	1

TABLE 1. For $T(3, n)$: the traceless character count N , the rank $2N + 1$ of the S-complex, the homology rank $\text{rank } I^\natural = \|\Delta_{T(3, n)}\|_1$, and the total rank of the S-complex differential (Proposition 5.4). The two ranks differ by 2 (bold) exactly for $n \equiv 4, 5 \pmod{6}$. The smallest case with nonzero differential is $8_{19} = T(3, 4)$, and the smallest odd case is $T(3, 5) = P(-2, 3, 5) = 10_{124}$.

For example, $T(3, 8)$ has $N = 5$ and chain and homology rank 11, so its S-complex differential vanishes, as [DS24b, Prop. 9.37] shows for the family $(3, 6k + 2)$ with $k = 1$; its homology rank exceeds $\det = 3$. For $T(3, 5) = P(-2, 3, 5)$ the homology has rank 7 rather than the chain rank 9 [DS24b, §9.3].

6. A NONZERO DIFFERENTIAL: $8_{19} = T(3, 4)$

We compare the instanton complex of the smallest case with nonzero S-complex differential, $8_{19} = T(3, 4)$, with the pillowcase complex computed by Hedden–Herald–Kirk.

6.1. The instanton complex. By Theorem 1.2 and Proposition 4.1, $T(3, 4)$ has three irreducible traceless characters, two non-dihedral (with $\cos \psi = \mp 1/\sqrt{3}$) and one dihedral (with $\cos \psi = 0$), together with one abelian character. On the double cover $\Sigma(2, 3, 4)$, whose first homology is $\mathbb{Z}/3$, the two non-dihedral characters lie over the unique irreducible flat connection, and the dihedral character lies over the nontrivial reducible (abelian) flat connection [DS24b, §9.1]. Here $\sigma = -6$, and [DS24a, Cor. 3] gives $C_* = \mathbb{Z}_{(1)}^2 \oplus \mathbb{Z}_{(3)}$, with the two non-dihedral characters in degree 1 and the dihedral one in degree 3 [DS24b, §9.4]. Thus $\tilde{C}(8_{19})$ has rank vector $(2, 2, 2, 1)$ in Daemi–Scaduto’s normalization, and $(2, 1, 2, 2)$ after the shift by $\sigma \equiv 2 \pmod{4}$ that matches the grading of $I^\natural$ [DS24b, Thm. 8.9], as in [PS17, Ex. 7.9]. The graded homology $I^\natural(8_{19})$ has rank vector $(2, 1, 1, 1)$ [HHK14, §12.2.3], [PS17, Ex. 7.8]. Comparing ranks grading by grading, a differential of degree -1 must have exactly one nonzero graded component, of rank one, from $I^\natural$ -grading 3 to $I^\natural$ -grading 2, that is, from $\text{gr}_{DS} = 1$ to $\text{gr}_{DS} = 0$. Daemi and Scaduto identify it: δ_1 is nonzero on C_1 , mapping the two characters lying over the irreducible connection to the reducible generator [DS24b, §9.4], and $h(T(3, 4)) = 1$ [DS24b, Thm. 1.14(iii)]. Poudel

and Saveliev state the same map in a cohomological convention, as a nontrivial map of degree $+1$ [PS17, Ex. 7.9].

6.2. The pillowcase complex. Hedden–Herald–Kirk compute the corresponding pillowcase complex [HHK18, §11.6]. They use a decomposition of $T(3, 4)$ along a Conway sphere bounding a ball in which the knot is a trivial tangle, with the parameters $(p, q, r, s) = (3, 4, 3, -2)$ of [HHK18, §10], where the integers r, s with $pr + qs = 1$ fix the decomposition; L_0 is the earring figure-eight of the trivial tangle, and L_1 is the perturbed character variety of the other tangle. The unperturbed variety is homeomorphic to the letter φ , a circle with an arc through it [HHK18, §11.6], [HHK14, Prop. 11.4], and under perturbation it resolves into an arc R_0 and a circle R_1 that winds twice around the pillowcase in the θ -direction, monotonically. Write $\text{gr}_{\mathcal{P}}$ for the $\mathbb{Z}/4$ grading of the pillowcase complex [HHK18, Def. 3.7]: on each component of L_1 it is a relative grading computed from Maslov indices of paths in L_0 and L_1 [HHK18, Lemma 3.10], a bigon lowers it by one [HHK18, Prop. 3.8], and on the component through r_+ it is made absolute by placing r_+ in grading $\sigma \bmod 4 = 2$ [HHK18, (19), §11.6]. The seven generators are the three points r_+, x_1^+, x_1^- where R_0 meets L_0 , with r_+ the perturbation of the abelian character, near the corner $(0, 0)$, and four generators on R_1 ; here $\text{gr}_{\mathcal{P}}(x_1^+) = 0$ and $\text{gr}_{\mathcal{P}}(x_1^-) = 3$, and R_1 contributes one generator in each grading. There is exactly one bigon, from x_1^- to r_+ , so $\partial x_1^- = r_+$, and the homology is

$$(1, 0, 0, 0)_{R_0} \oplus (1, 1, 1, 1)_{R_1} = (2, 1, 1, 1),$$

in agreement with $I^{\natural}(8_{19})$. The bigon has one corner at r_+ , near $(0, 0)$, and the other at x_1^- , near the diagonal [HHK18, Fig. 21].

6.3. Comparison. The two computations exhibit a rank-one map of the same degree, from grading 3 to grading 2, ending at the generator of the abelian character. They concern different complexes: no chain-level identification of the pillowcase complex with $\tilde{C}$ is known in general, and the conjectures recalled in Section 1.1 concern homology. A chain-level conjecture over $\mathbb{F}_2$ for torus knots is formulated in [Wue26b, Conj. 1.5], and for $T(3, n)$ with $n \geq 4$ and perturbations in the cone of [Wue26c, Def. 5.5] it is proved in [Wue26c, Thm. 1.3]. In the S-complex structure on the pillowcase complex used there [Wue26b, §1.3], with r_+ as the reducible generator and x_i^-, x_i^+ as the two copies of an irreducible generator, the differential is a single component of type δ_1 , of the same rank as $\tilde{d}$; for $n \equiv 4, 5 \pmod{6}$ it is given by one bigon, ending at r_+ . The pillowcase differential also depends on the perturbation: for a larger perturbation parameter the two generators joined by the bigon cancel and the perturbed complex is perfect [HHK18, §11.10, Prop. 11.1]. By contrast, $\delta_1 \neq 0$ in every S-complex representing $\tilde{C}(T(3, 4))$, because $h(T(3, 4)) = 1$ [DS24b, Prop. 4.15]. The same kind of bigon, from x_1^- to r_+ , occurs in Hedden–Herald–Kirk’s computation for $T(3, 5)$ [HHK18, §11.7], whose double branched cover has no nontrivial reducible flat connection, so the bigon is not tied to the nontrivial reducible of $\Sigma(2, 3, 4)$; for two-bridge knots L_1 is straight and there is no bigon (Section 3.2). For 8_{19} the chain rank 7 exceeds $\det = 3$ by the four generators of the two non-dihedral characters, and the differential reduces it to $\text{rank } I^{\natural}(8_{19}) = 5$.

7. OPEN QUESTIONS

On the instanton side, the invariants considered here are determined for every $T(3, n)$: the rank of the S-complex differential is zero or one according to $n \bmod 6$, and in the rank-one cases its component δ_1 is nonzero (Theorem 1.3). On the pillowcase side, for the decomposition of [HHK18, §10] with the integers (r, s) used there for $T(3, 4)$, $T(3, 5)$ and $T(3, 7)$, and for sufficiently small perturbations in the closed cone of [Wue26c, Def. 5.5], the pillowcase Floer homology of every $T(3, n)$ with $n \geq 4$ has the $\mathbb{Z}/4$ -graded dimensions of $I^{\natural}$, so the conclusion

of [HHK18, Conj. 6.5] holds for this family [Wue26c, Cor. 1.2]; after its grading is shifted by σ and its closed components are suitably graded, the pillowcase complex is S-chain homotopy equivalent to $\tilde{C}(T(3, n)) \otimes \mathbb{F}_2$ [Wue26c, Thm. 1.3]. Hedden–Herald–Kirk computed pillowcase Floer homology with zero bounding cochains, of the correct rank, for $T(3, 4)$, $T(3, 5)$ and $T(3, 7)$, and for the $(4, 5)$ -torus knot [HHK18, §§11.2, 11.6–11.8]; the evidence that other decompositions need nonzero cochains is recalled in Remark 3.2. The companion paper [Wue26a] determines $I^\natural(P(-2, 3, q); \mathbb{Z})$ for the pretzel knots $P(-2, 3, q)$, a family containing $T(3, 4) = P(-2, 3, 3)$ and $T(3, 5) = P(-2, 3, 5)$, proves [HHK18, Conj. 6.5] for a decomposition of each $P(-2, 3, q)$ with $q \geq 7$, and studies Maurer–Cartan deformations of explicit immersed curves for Smith’s decomposition in an algebraic model of the wrapped Fukaya category of the pillowcase with its four corners removed.

We list three open questions; Conjecture D of [CHKK22] is recalled in Section 1.1.

- (1) Construct the bounding cochains of [CHKK22, Conj. D] for decompositions in which zero cochains give the wrong rank, such as those recalled in Remark 3.2, and prove that they restore the rank of $I^\natural$. For the second decomposition of $T(3, 4)$ in that remark, a cochain with this property is found numerically in [Wue26c, §7].
- (2) Extend the chain-level identification over $\mathbb{F}_2$ from the knots $T(3, n)$ to all torus knots, as conjectured in [Wue26b, Conj. 1.5].
- (3) Identify the Maurer–Cartan deformations of [Wue26a] with the bounding cochains of [CHKK22].

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